The Capacity Region of the Binary Symmetric Z-Interference Channel

[Author list TBD]

Abstract

We give a complete, self-contained closed-form characterization of the capacity region of the two-user binary symmetric Z-interference channel $Y_1 = X_1 \oplus X_2 \oplus N_1$, $Y_2 = X_2 \oplus N_2$, with independent noises $N_i \sim \mathrm{Bern}(p_i)$, $p_1, p_2 \in (0, \frac{1}{2})$, for all parameters. When the interfered receiver is the less noisy one ($p_1 \leq p_2$), the region is the polytope $\{R_1 + R_2 \leq 1 - h(p_1),\ R_2 \leq 1 - h(p_2)\}$; in the weak regime ($p_1 > p_2$) the region coincides with that of the degraded binary symmetric broadcast channel with parameters (p_1, p_2), treating interference as noise with an optimized input bias is capacity-achieving, rate splitting is unnecessary, and the sum capacity $1 - h(p_2)$ is attained only with the interfering user silent. The weak-regime converse is an eight-line application of Mrs. Gerber's Lemma, and achievability follows from a two-line algebraic identity. The ingredients are classical — the strong regime instantiates strong-interference machinery, and the weak regime is capacity-region equivalent to Benzel's discrete additive degraded interference channel — but we are not aware of the closed form for all parameters, or of this short a derivation, appearing in one place.

I. Introduction

Interference, not noise, is the binding constraint in multiuser wireless systems, and the two-user interference channel is its canonical model. The one-sided (or Z-) interference channel, in which only one receiver suffers interference, is the minimal such model: it isolates a single interference link, and every phenomenon of interference management — rate splitting, joint decoding, treating interference as noise (TIN) — already appears in it. For the Gaussian Z-interference channel the capacity region in the weak-interference regime remains a long-standing open problem despite decades of effort [5]. It is therefore natural to ask for the simplest noisy, discrete instance: the binary symmetric Z-interference channel (BS-ZIC),

$$Y_1 = X_1 \oplus X_2 \oplus N_1, \qquad Y_2 = X_2 \oplus N_2, \tag{1}$$

with $N_1 \sim \mathrm{Bern}(p_1)$, $N_2 \sim \mathrm{Bern}(p_2)$ independent of each other and of the inputs, $p_1, p_2 \in (0, \frac{1}{2})$. Receiver i decodes only message i; the encoders are independent. This paper records the complete capacity region of (1), in closed form, for all parameter values — together with proofs short enough to be taught in a first course.

The result is significant less for its difficulty than for what it crystallizes. First, it exhibits the sharpest known contrast between the binary and Gaussian one-sided interference problems: the binary weak regime closes in eight lines because Mrs. Gerber's Lemma [1] converts noise-degradation into a converse exactly, whereas the Gaussian analogue (via the entropy power inequality) famously does not close the corresponding gap. Second, it gives a complete, extreme answer to the TIN question for this channel: treating interference as noise with a single biased input is exactly optimal throughout the weak regime — no rate splitting, no auxiliary time sharing — a discrete counterpart to the TIN-optimality program of [10]. Third, the weak-regime region coincides, boundary point by boundary point, with the degraded binary symmetric broadcast channel region of Wyner–Ziv [1]: the interfering codeword plays precisely the role of a broadcast "cloud center," making the BS-ZIC a clean pedagogical bridge between interference and broadcast theory.

A. Related work

The strong-interference program originates with Carleial [2] and Sato [3] and was completed for the discrete memoryless case by Costa and El Gamal [4]; textbook treatments of the one-sided case appear in El Gamal and Kim [9, Ch. 6]. Our strong-regime result (Theorem 1(i)) is an instantiation of that machinery. For the weak regime, Benzel [6] determined the capacity region of a class of discrete additive degraded interference channels; Liu and Ulukus [7] extended the class and its broadcast-channel correspondence; and Liu and Goldsmith [8] characterized the capacity region of a class of Z-interference channels containing (1) in single-letter (auxiliary-variable) form, observing the capacity-region equivalence between the Z-channel form and Benzel's two-output additive model. The closed form of Theorem 1(ii) is thus classical in substance: our contribution is the explicit assembled statement for all (p_1, p_2), the direct Mrs. Gerber converse (which bypasses the broadcast-channel correspondence entirely), the two-line achievability identity, and the uniqueness of the sum-optimal point. For the Gaussian Z-interference channel, by contrast, the weak-interference capacity region remains open [5]; the best known results characterize sum capacity only in restricted regimes.

II. Model and Main Result

Throughout, h is the binary entropy function, $h^{-1} : [0,1] \to [0, \frac{1}{2}]$ its inverse on $[0, \frac{1}{2}]$, and $a \star b = a(1-b) + b(1-a)$ the binary convolution. Rates are in bits per channel use; a rate pair is achievable in the standard sense of vanishing average error probability.

Theorem 1: The capacity region of the BS-ZIC (1) is:

(i) if $p_1 \le p_2$ (strong-to-moderate interference):

$$\big\{(R_1, R_2) \ge 0 : \;\; R_1 + R_2 \le 1 - h(p_1), \quad R_2 \le 1 - h(p_2)\big\};$$

(ii) if $p_1 > p_2$ (weak interference), with $q = \dfrac{p_1 - p_2}{1 - 2p_2} \in (0, \frac{1}{2})$:

$$\big\{(R_1, R_2) \ge 0 : \;\; R_2 \le 1 - h(p_2), \quad R_1 \le 1 - h\big(q \star h^{-1}(h(p_2) + R_2)\big)\big\}.$$

In the weak regime the sum capacity is $1 - h(p_2)$, attained only at $(0,\, 1 - h(p_2))$.

Part (ii) is exactly the capacity region of the degraded binary symmetric broadcast channel with noise parameters (p_1, p_2) [1], and its boundary is traced by treating interference as noise with a biased input for user 2 and a uniform input for user 1.

III. Strong-to-Moderate Interference

a) Converse: Since $p_1 \le p_2$, write $p_2 = p_1 \star p_0$ with $p_0 \in [0, \frac{1}{2})$ and couple $N_2^n = N_1^n \oplus N_0^n$, where $N_0 \sim \mathrm{Bern}(p_0)$ is i.i.d. and independent of everything else; this preserves the joint law of (X_2^n, Y_2^n). Data processing then gives

$$I(X_2^n; Y_2^n) \;\le\; I(X_2^n; X_2^n \oplus N_1^n) \;=\; I(X_2^n; Y_1^n \mid X_1^n),$$

the identity because, given X_1^n, the map $Y_1^n \leftrightarrow X_2^n \oplus N_1^n$ is a bijection, and the encoders are independent. With Fano's inequality,

$$n(R_1 + R_2) \le I(X_1^n; Y_1^n) + I(X_2^n; Y_1^n \mid X_1^n) + n\epsilon_n = I(X_1^n, X_2^n; Y_1^n) + n\epsilon_n \le n\big(1 - h(p_1)\big) + n\epsilon_n,$$

using $H(Y_1^n) \le n$ and $H(Y_1^n \mid X_1^n, X_2^n) = nh(p_1)$; and $nR_2 \le n(1 - h(p_2)) + n\epsilon_n$ is the point-to-point bound.

b) Achievability: Take both inputs i.i.d. uniform. Receiver 1 jointly decodes both messages over the mod-2 multiple-access channel $Y_1 = (X_1 \oplus X_2) \oplus N_1$, which succeeds iff $R_1, R_2, R_1 + R_2 \le 1 - h(p_1)$; receiver 2 requires $R_2 \le 1 - h(p_2)$. Since $p_1 \le p_2$, the constraint $R_2 \le 1 - h(p_1)$ is implied by $R_2 \le 1 - h(p_2)$, and the achievable region equals the converse region. This is the classical strong-interference argument [3], [4], [9] instantiated at (1). ■

IV. Weak Interference

a) Converse: Now $p_1 = p_2 \star q$ with $q = (p_1 - p_2)/(1-2p_2) \in (0, \frac{1}{2})$; couple $N_1^n = N_2^n \oplus Q^n$ with $Q \sim \mathrm{Bern}(q)$ i.i.d. independent, preserving the law of (X_1^n, X_2^n, Y_1^n). Let $V^n = X_2^n \oplus N_2^n$, which is distributed as Y_2^n. Fano's inequality at receiver 2 gives

$$\tfrac{1}{n} H(V^n) \;\geq\; h(p_2) + R_2 - \epsilon_n.$$

Mrs. Gerber's Lemma [1] states that for any binary V^n and independent i.i.d. $Q^n \sim \mathrm{Bern}(q)^n$, $\frac{1}{n}H(V^n \oplus Q^n) \geq h\big(q \star h^{-1}(\frac{1}{n}H(V^n))\big)$. Therefore

$$nR_1 \;\leq\; n - H(Y_1^n \mid X_1^n) + n\epsilon_n \;=\; n - H(V^n \oplus Q^n) + n\epsilon_n \;\leq\; n\Big[1 - h\big(q \star h^{-1}(h(p_2) + R_2 - \epsilon_n)\big)\Big] + n\epsilon_n,$$

using $H(Y_1^n \mid X_1^n) = H(X_2^n \oplus N_1^n) = H(V^n \oplus Q^n)$ (independent encoders) and the monotonicity of $h(q \star h^{-1}(\cdot))$. Letting $n \to \infty$ yields the stated bound.

b) Achievability: Treat interference as noise with a biased input: let $X_2 \sim \mathrm{Bern}(\pi)$ i.i.d. carry user 2's message and $X_1 \sim \mathrm{Bern}(\frac{1}{2})$ user 1's. Receiver 2 achieves $R_2 = I(X_2; Y_2) = h(\pi \star p_2) - h(p_2)$. Receiver 1 sees $Y_1 = X_1 \oplus Z$ with $Z \sim \mathrm{Bern}(\pi \star p_1)$, so $R_1 = I(X_1; Y_1) = 1 - h(\pi \star p_1)$. Substituting $s = \pi \star p_2$, so that $h(s) = h(p_2) + R_2$, and using $p_1 = p_2 \star q$, the associativity and commutativity of $\star$ give $\pi \star p_1 = (\pi \star p_2) \star q = s \star q$, hence

$$R_1 \;=\; 1 - h\big(q \star h^{-1}(h(p_2) + R_2)\big) :$$

the inner bound traces the converse curve exactly as π runs over $[0, \frac{1}{2}]$. Lower rates follow by coding below the two point-to-point mutual informations.

c) Uniqueness of the sum-optimal point: Along the boundary, $R_1 + R_2 = 1 - h(p_2) + h(s) - h(q \star s)$ with $s \in [p_2, \frac{1}{2}]$. For $s < \frac{1}{2}$ and $q > 0$ one has $s < q \star s < \frac{1}{2}$, hence $h(s) < h(q \star s)$ and the sum is strictly below $1 - h(p_2)$; equality holds only at $s = \frac{1}{2}$, i.e. at the corner $(0, 1 - h(p_2))$: any sum-optimal operation silences the interfering user. ∎

V. Discussion

a) Broadcast equivalence: In the weak regime, receiver 1 observes user 2's codeword through strictly more noise than receiver 2 does; the mirror coupling $N_1 \stackrel{d}{=} N_2 \oplus \mathrm{Bern}(q)$ makes this a literal degradation, and user 2's codeword plays the role of the cloud center in a degraded broadcast channel. Mrs. Gerber's Lemma is precisely the tool that makes the Wyner–Ziv broadcast converse work, and it transplants verbatim. After relabeling, the channel is capacity-region equivalent to Benzel's discrete additive degraded interference channel [6] (the equivalence is stated by Liu–Goldsmith [8]; Benzel's literal two-output additive law differs from (1)), and the channel satisfies both defining conditions of the Liu–Goldsmith class [8] exactly and in n-letter form — mod-2 additivity makes $H(Y_1^n \mid X_1^n{=}x_1^n)$ invariant in x_1^n, and the uniform input maximizes the interfered receiver's output entropy irrespective of the interference law — so Theorem 1(ii) can equally be read as a closed-form evaluation of their auxiliary characterization, with a degenerate auxiliary.

b) No rate splitting anywhere: In the strong regime the optimal scheme decodes the interference completely; in the weak regime it ignores the interference completely. The Han–Kobayashi partial-decoding machinery, generically necessary for interference channels, is nowhere needed for (1) — an extreme, exactly provable instance of the TIN-optimality phenomenon studied in the Gaussian setting in [10].

c) The Gaussian contrast: The binary weak regime closes because Mrs. Gerber's Lemma gives the exact extremal tradeoff between input entropy and output entropy across a BSC. The Gaussian analogue of that role is played by the entropy power inequality, which is not tight for the interference geometry, and the Gaussian Z-interference channel's weak regime remains open [5]. The BS-ZIC thus marks, in the simplest possible setting, exactly where the discrete theory outruns the Gaussian one.

Contribution and AI-Use Statement

d) Initial generation: The strong-interference theorem was first proved by a Gemini-based AI agent; the weak-regime converse, achievability identity, uniqueness argument, and this manuscript were produced by Claude Fable 5 (Anthropic), all within a public open-problems repository.

e) Independent audit: GPT-5 Codex (OpenAI) audited both regimes, identified the correct classical attribution (Benzel; Liu–Goldsmith equivalence), and contributed the strict sum-uniqueness argument's final form.

f) Human responsibility: The human maintainer supervised the process, reviewed the mathematics, and is responsible for the content. The results are classical in substance; this note claims presentational, not mathematical, novelty.

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